🧠 Aging Heap Allocator (AHA)

Language: ANSI C
Architecture: Memory Lifecycle Simulation System
Execution Model: Deterministic Aging-Based Allocation
Version: 1.0

📌 Executive Summary

The Aging Heap Allocator (AHA) is a structured memory lifecycle simulation system developed in ANSI C.

The system models-controlled memory allocation and automated deallocation based on block aging policies.

It integrates:

• Simulated heap block allocation
• Memory aging cycles
• Expiration-based deallocation
• Lifecycle policy evaluation

AHA demonstrates core principles of memory management strategy design and aging-based garbage collection models.

🏗️ System Architecture

AHA follows a layered logical architecture:

• Allocation Layer
• Aging Mechanism
• Expiration Policy Engine
• Lifecycle Controller

Although simplified, it models behavior similar to generational garbage collectors and aging-based memory schedulers.

🧩 Memory Allocation Layer

Each memory block contains:

• Block ID
• Allocated Size
• Age Counter

Functional Capabilities:

• Initialize memory block
• Assign block identity
• Set initial age to zero

This simulates heap block creation.

⏳ Aging Mechanism

The system increments the age counter during each cycle.

Operational Logic:

• Each execution cycle increases block age
• Age accumulation simulates memory residency time

Aging simulates runtime exposure and memory retention duration.

🗑️ Expiration Policy Engine

The expiration mechanism determines when a block should be deallocated.

Deallocation Condition:

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Deallocate if Age > Threshold
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Default Threshold:

5 cycles

Purpose:

• Simulate memory pressure control
• Demonstrate lifecycle-based cleanup
• Prevent indefinite retention

📊 Lifecycle Evaluation Model

This model allows:

• Controlled memory lifetime
• Expiration simulation
• Policy-based memory recycling

It mirrors simplified generational garbage collection strategies used in managed runtime systems.

🧠 Engineering Highlights

• Memory lifecycle abstraction
• Deterministic aging simulation
• Expiration-based deallocation policy
• Structured heap modeling
• Controlled memory behavior representation

🎯 Project Significance

AHA demonstrates principles of:

• Memory management strategy design
• Heap lifecycle modeling
• Resource aging policy simulation
• Garbage collection concepts

It reflects foundational systems-level memory engineering concepts.

🏁 Conclusion

The Aging Heap Allocator (AHA) is a deterministic memory lifecycle simulation platform implemented in ANSI C.

It integrates:

Heap Allocation
Aging Simulation
Expiration Policy
Controlled Deallocation

All within a structured and predictable runtime environment.